

0 6 . 1

Define genome and proteome.

[2 marks]

Genome _____

Proteome _____

The classification system used in the early 20th century grouped different species of bacteria according to the position and shape of flagella on bacterial cells and by the number of flagella per cell. These were observed using an optical microscope. Each species of bacterium has a characteristic cell shape and arrangement of flagella. These characteristics may be shared with other species within a genus. Flagella are fragile, difficult to stain and may extend from the cell at any angle.

0 6 . 2

Consider the accuracy and limitations of the early classification of bacteria using the arrangement of flagella.

[3 marks]

0 6 . 3

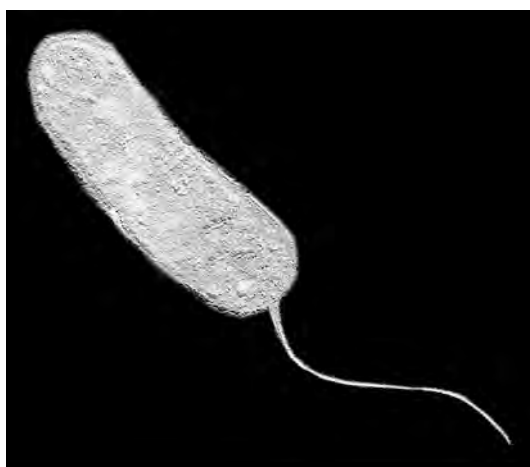
Suggest why several bacterial species have been renamed in recent years.

[1 mark]

0 6 . 4

Figure 6 shows an image from an optical microscope of a single bacterial cell.

Figure 6



This bacterial cell is $2.3\text{ }\mu\text{m}$ long (excluding the flagellum).

Calculate the magnification of this image.

Show your working.

[2 marks]

Magnification $\times$ _____

Turn over for the next question

Turn over ►

